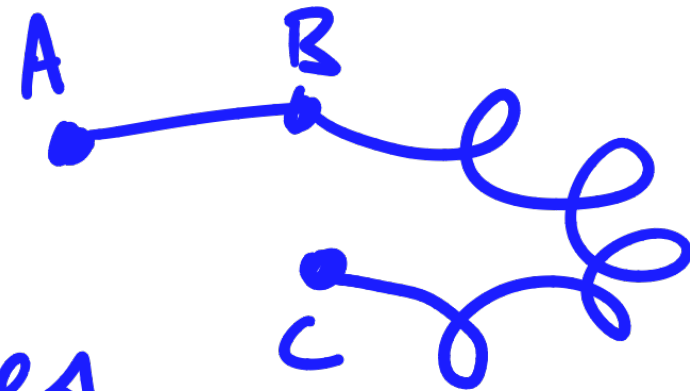


Lecture Notes: Graph Theory, Intro

Math F113X Math and Society

Vertices and edges

vertex (plural: vertices): Dots/circles that represent things



edge: connections between vertices

Pairs of vertices = edges

Edges are AB
AC

Building the model

situation	vertices are...	edges are...
The <u>road system</u> connecting some <u>towns</u> in Alaska.	towns	roads
The streets in a neighborhood that a snowplow has to clear after a storm.	intersections	roads

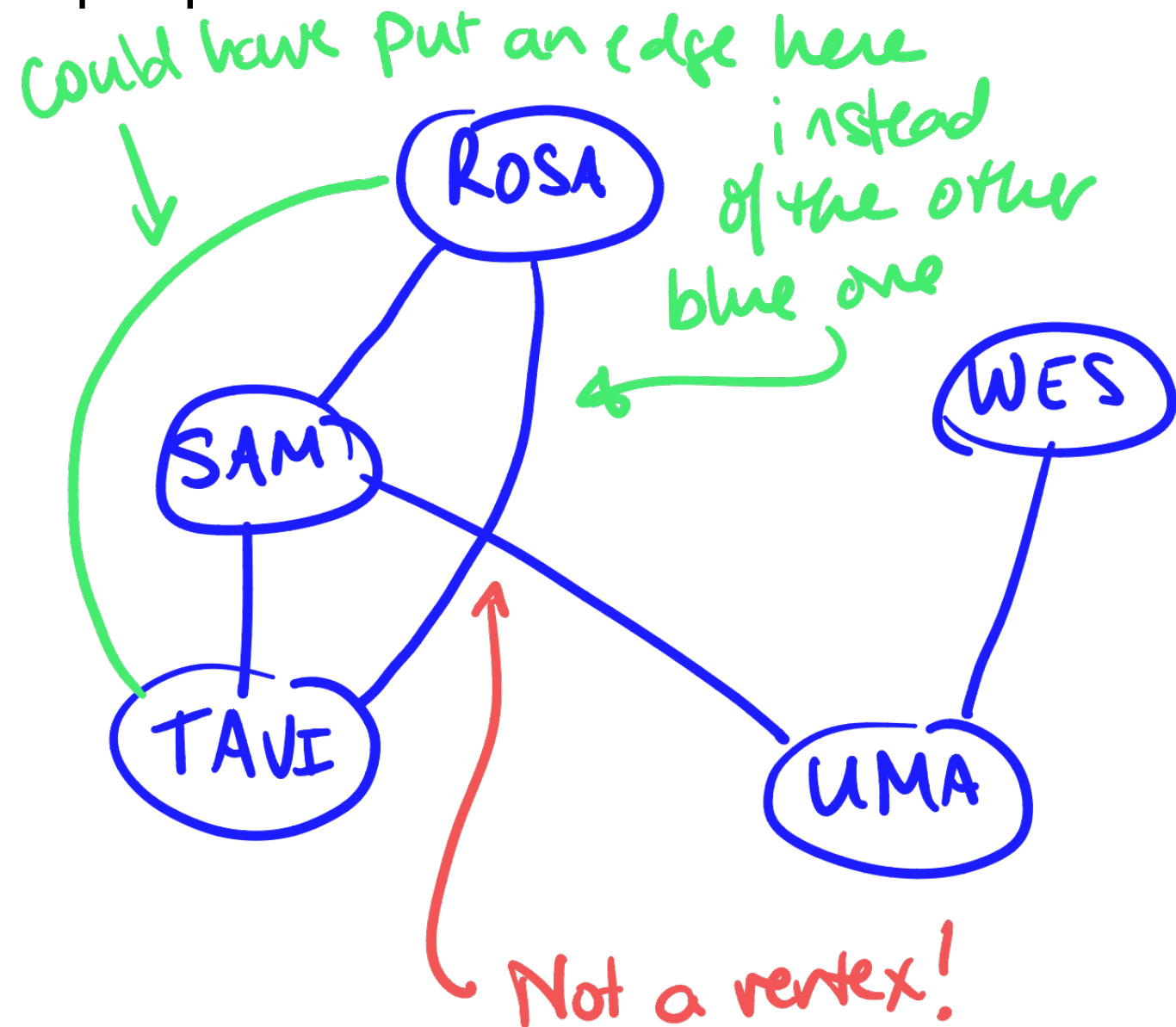
Building the model

situation	vertices are...	edges are...
A group of people, and who is friends with whom.	People	connections between friends
Classes offered at this university, and the students that take them.	Classes	edge if one student taking both classes
	Students	edge if in same class

Draw a graph

We have information about five people and who is friends with whom.
Draw a graph.

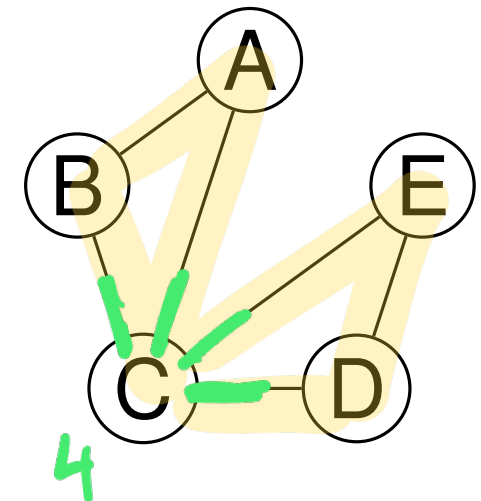
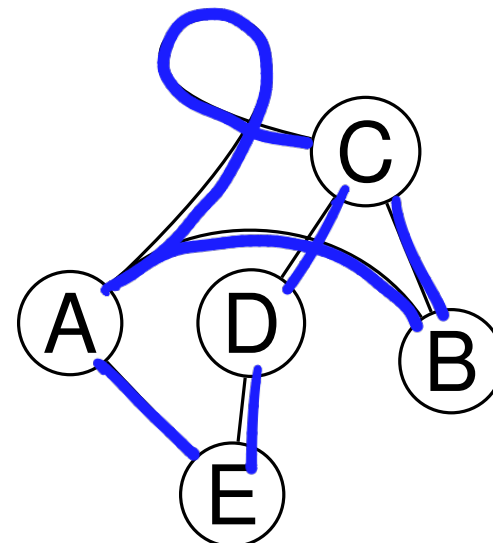
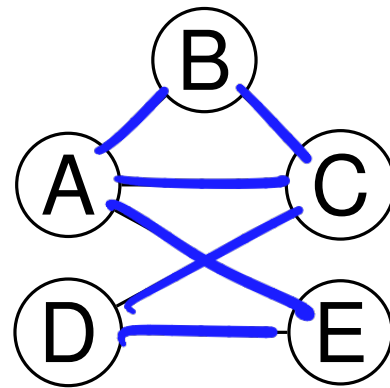
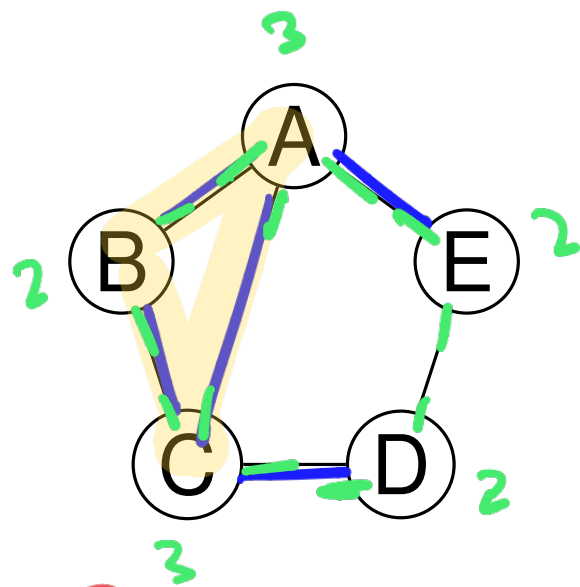
person	friends with
<u>Rosa</u>	<u>Sam</u> , <u>Tavi</u>
<u>Sam</u>	<u>Rosa</u> , <u>Tavi</u> , <u>Uma</u>
<u>Tavi</u>	<u>Rosa</u> , <u>Sam</u>
<u>Uma</u>	<u>Sam</u> , <u>Wes</u>
<u>Wes</u>	<u>Uma</u>



Are these all the same graph?

How can we tell? (What does "same" mean?)

All have 5 vtx
& 6 edges!



Same means: equivalent vertices & edges
different: Something that doesn't match!

AB BC
AC CD
AE DE

AB BC
AC CD
AE DE

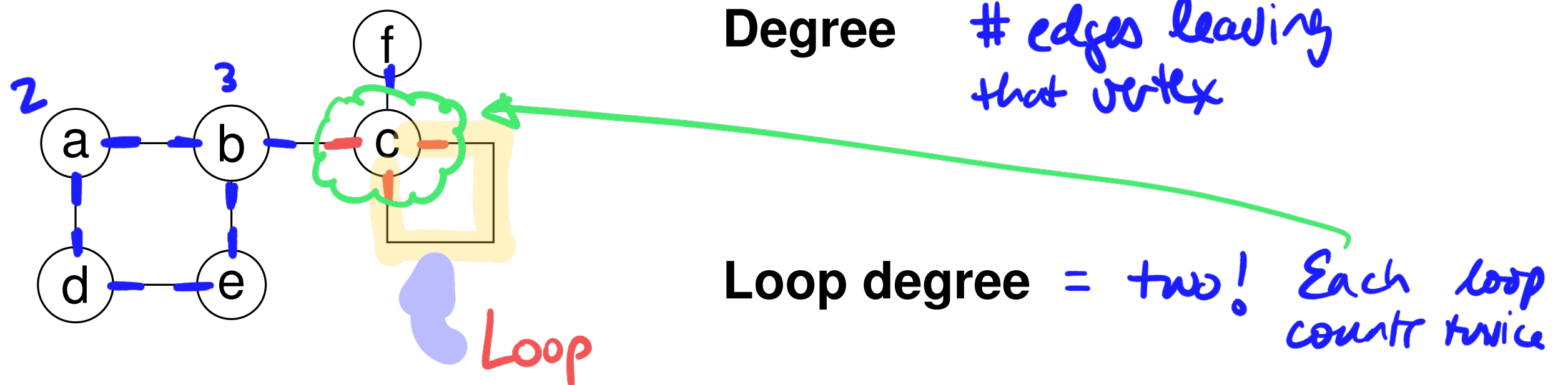
AB BC
AC CD
AE DE

SAME!

Not the same!
2 Δ s here,
but other only
has one!

Degree

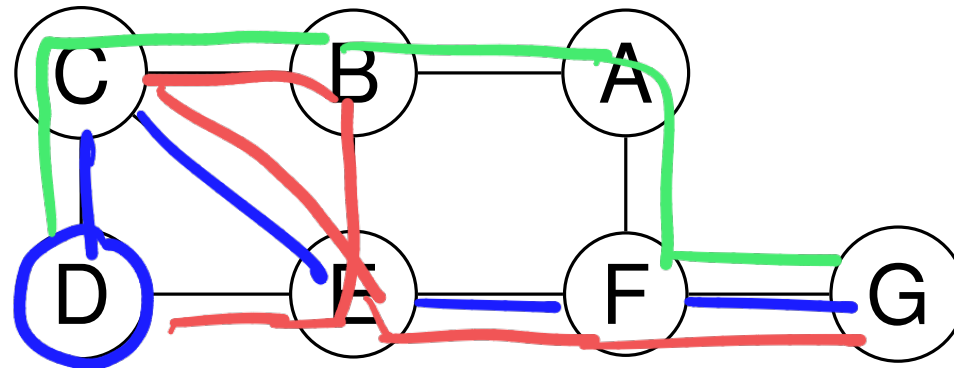
Below is a neighborhood: each vertex is an intersection, each edge is a street. How busy is each intersection?



degree of a : 2 b : 3 c : 5 d : 2 e : 2 f : 1

Getting around: **path**

Below is a road network: each vertex is a town, each edge is a road.



Can I get from here to there?

path: *Sequence of vertices & edges*

What's the shortest path?

What's the longest path?

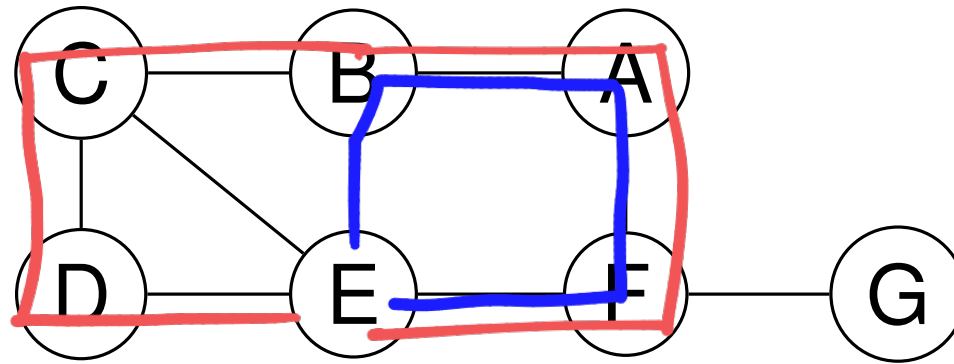
Find a path from D to G :

$DCEFG$

$CEBCFG$

$DCBAFG$

Getting around: **circuit**



Can I leave home and get back?

circuit: A path that starts and stops at the same vertex

Find a circuit starting at E:

E B A F E
E D C B A F E

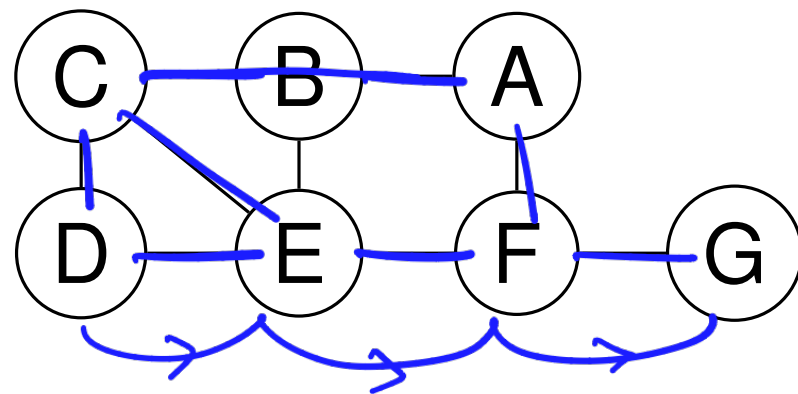
Getting around: **connected**

Is everything reachable at all?

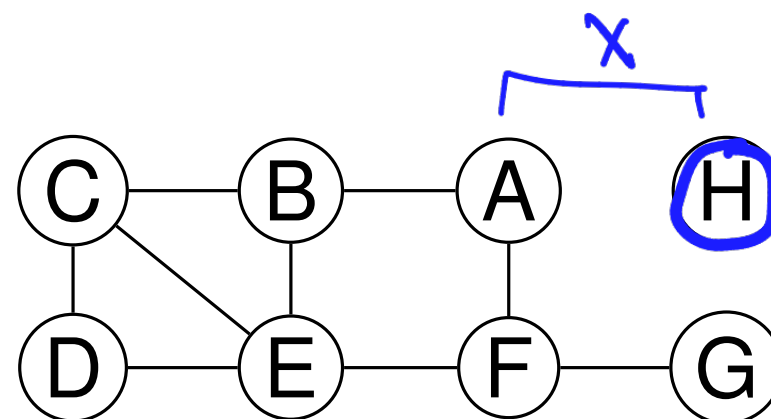
connected: *Is there a path between any pair of vertices?*

Not connected: you can find a pair that is not connected by a path

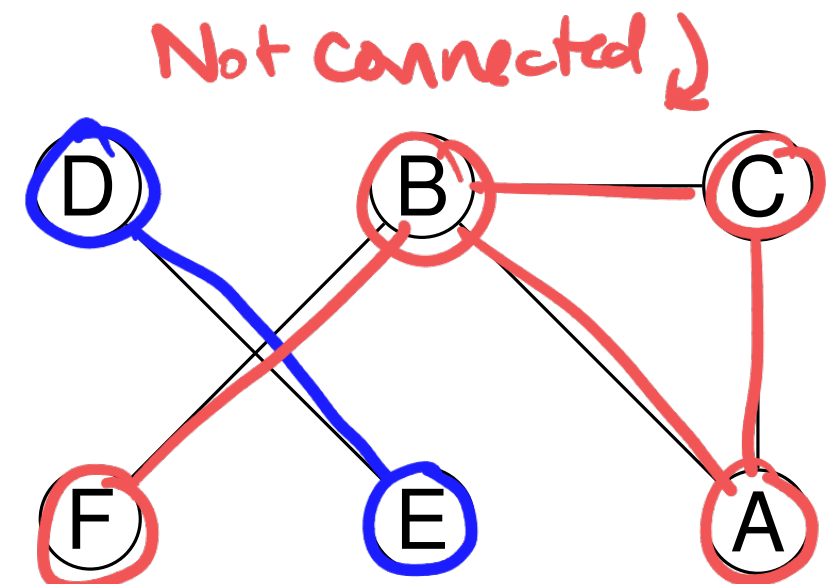
Which graphs are connected?



Connected



Not connected



Not connected ↓

*two connected
components*